

Engineering Department Doc, General Developer 1 Final Handoff Summary

Soccer TIPS: Tactical Intelligence & Performance System

Final closeout summary for General Developer 1

1. Document Purpose

This document is the final closeout summary for General Developer 1. It summarizes the implementation work completed across Layer 1 ingestion infrastructure, Layer 2 feature-engineering foundation, and Layer 2 football sanity validation.

Future developers should use this as a navigation document. It is not a replacement for the detailed engineering handoff documents. The detailed documents remain the source for implementation specifics, validated dataset behavior, run commands, QA details, and scope boundaries.

2. Documents Future Developers Must Read

Item	Current status / meaning
AIB Project System Architecture / Workflow Specification	Explains the full layered system, project philosophy, AI boundaries, and how each layer connects to the broader Soccer TIPS workflow.
Engineering Department Doc, Validation & Data Ingestion Handoff Summary	Explains why the Layer 1 ingestion rules exist, including SkillCorner and Metrica dataset behavior, parser rules, coordinate handling, missing-file behavior, and validation findings.
Engineering Department Doc, Layer 1_v0.1 General Developer Handoff Notes	Explains what was built for reusable Layer 1 ingestion, how canonical processed inputs are created, how batch processing works, and what warnings are expected.
Engineering Department Doc, Layer 2_v0.2 General Developer Handoff Notes	Explains what was built for Layer 2 v0.1, how the feature tables are generated, how QA is performed, and how Layer 2 v0.2 football sanity validation was completed.
Layer 2 football sanity validation section/addendum	Documents the football sanity validation workflow, phase-specific metric checks, validation plots, reasonableness checks, and the team_width caution that remains before stronger tactical claims.

3. Work Completed by General Developer 1

Layer 1 work

- Built provider-specific ingestion loaders for SkillCorner and Metrica.
- Created shared canonical player and ball tracking outputs for downstream use.
- Preserved raw provider coordinates and generated metric coordinates separately where needed.
- Implemented validation report creation and controlled Layer 1 batch processing.
- Added Layer 1 QA inspection with schema compliance checks.
- Supported SkillCorner Open Data and Metrica Sample_Game_1 / Sample_Game_2. Metrica Sample_Game_3 remains intentionally out of scope.

Layer 2 work

- Created the src/soccer_tips/features/ package.
- Implemented SkillCorner phase-context linking using phases_of_play.csv.
- Computed frame-level team shape metrics from Layer 1 canonical tracking outputs.
- Aggregated frame-level metrics into phase-segment summaries and match tactical profiles.

- Built Layer 2 tests, build runner, QA utility, and batch support across all processed SkillCorner matches.
- Built Layer 2 v0.2 football sanity validation outputs, including phase metric summaries, structured JSON summary, and validation plots.

4. Final Accepted Project State

- Layer 1 v0.1 is accepted as reusable ingestion infrastructure.
- Layer 2 v0.1 is accepted as the feature-engineering foundation.
- Layer 2 v0.2 football sanity validation is completed with final QA status warning, not failure.
- The remaining Layer 2 v0.2 warning is team_width_pitch_bound. It should be reviewed before team_width is used for stronger tactical claims.
- SkillCorner is the primary provider for phase-aware Layer 2 work because it has validated phases_of_play data.
- Metricka is supported for Layer 1 ingestion but is not currently used for phase-aware Layer 2 analysis.
- No Layer 3 modeling, clustering, tactical identity modeling, AI interpretation, dashboards, reports, or recommendation systems have been added yet.

5. Current Code Structure

Item	Current status / meaning
src/soccer_tips/data/	Layer 1 ingestion, provider-specific loaders, coordinate normalization, validation utilities, Layer 1 build runner, and Layer 1 QA inspection.
src/soccer_tips/features/	Layer 2 feature schemas, SkillCorner phase-context linking, orientation placeholder module, team shape metrics, aggregation, Layer 2 build runner, Layer 2 QA, football sanity validation, and validation plotting.
tests/	Lightweight test scripts for Layer 1 ingestion, Layer 2 feature engineering, and Layer 2 football sanity helpers.
data/processed/	Local generated outputs from Layer 1 and Layer 2. This should remain reproducible output data, not manually edited source material.
data/processed/features/	Layer 2 feature outputs, QA summaries, and football sanity summaries.
outputs/figures/layer2_sanity/	Validation-only Layer 2 football sanity figures. These are sanity plots, not dashboards or final reports.

6. Important Run Commands

Open PowerShell from the project root:

```
cd "C:\Users\amsch\. Personal Work\Projects\Soccer TIPS"
```

Standard setup commands:

```
.\env\Scripts\Activate
pip install -r requirements.txt
$env:PYTHONPATH="src"
```

Item	Current status / meaning
Run Layer 1 tests	python tests/test_layer1_ingestion.py
Run Layer 1 batch	python -m soccer_tips.data.build_layer1 --provider all --batch
Run Layer 1 QA	python -m soccer_tips.data.qa_layer1_outputs
Run Layer 2 tests	python tests/test_layer2_features.py
Run Layer 2 batch	python -m soccer_tips.features.build_layer2 --provider skillcorner --batch
Run Layer 2 QA	python -m soccer_tips.features.qa_layer2_outputs
Run football sanity tests	python tests/test_football_sanity.py
Run Layer 2 football sanity validation	python -m soccer_tips.features.qa_football_sanity

7. Key Technical Decisions Made

- Use provider-specific loading logic with shared canonical outputs.
- Use SkillCorner phases as the trusted phase-context layer for current phase-aware Layer 2 work.
- Preserve raw provider coordinates and metric coordinates separately.
- Keep orientation normalization intentionally inactive until it is validated and approved.
- Preserve detection and extrapolation flags for quality diagnostics and future confidence filtering.
- Treat missing SkillCorner dynamic events as a warning, not a failure.
- Keep Metrica Sample_Game_3 out of current scope because it uses a different internal format.
- Start Layer 2 with simple, interpretable team shape metrics before deeper tactical interpretation.
- Do not add Layer 3 modeling or AI interpretation before descriptive tactical profiles exist.

8. Known Warnings and Limitations

- SkillCorner match 1953632 is missing optional dynamic_events.csv. This is expected and non-fatal.
- Metrica Sample_Game_1 contains one known kickoff initialization event range issue. This is expected and non-fatal.
- Orientation normalization is inactive. Current analysis uses Layer 1 metric coordinates.
- Layer 2 v0.2 football sanity QA finished with status warning because team_width_pitch_bound requires review.
- centroid_x should not be used for strong directional claims until attacking direction/orientation is validated.
- There are no final tactical identity claims yet.
- There is no clustering yet.
- There is no AI interpretation yet.
- There is no dashboard or reporting layer yet.
- Current Layer 2 outputs are feature and validation outputs, not final scouting conclusions.

9. What Future Developers Should NOT Redo

- Do not redo Layer 1 ingestion from scratch unless a validated defect requires it.
- Do not change canonical schemas casually.
- Do not rebuild SkillCorner or Metrica loaders without a clear reason.
- Do not treat raw tracking possession as authoritative without validation.
- Do not activate orientation normalization casually.
- Do not jump straight to clustering.
- Do not use Layer 2 metrics for final tactical claims without sanity validation and Lead Developer review.
- Do not create AI interpretation before Layer 3 descriptive outputs exist.

10. Recommended Next Developer Role

The next active developer should be a Data Analyst Specialist.

This role should focus on turning accepted Layer 2 outputs into descriptive tactical profiles, not machine-learning-first modeling. The next developer should work on:

- descriptive tactical profiles;
- tactical fingerprint schema design;
- stability analysis;
- team comparison;
- similarity scoring;
- optional clustering only later, after descriptive profiling is sound.

The next developer should not start with machine learning or AI interpretation.

11. Recommended Next Technical Phase

Recommended next phase: Layer 3 v0.1 - Descriptive Tactical Profiling.

Suggested goals:

- define the first canonical tactical fingerprint schema;
- summarize team shape tendencies by phase;
- create interpretable match/team profiles;
- identify stable versus unstable tactical behaviors;
- compare teams using simple descriptive statistics;
- prepare structured outputs for future AI interpretation.

12. Final General Developer 1 Closeout Note

The project has moved from raw dataset exploration to reusable ingestion and feature infrastructure. Future work should build on the existing modules rather than replace them. The next development phase should preserve the same engineering discipline: validate first, document decisions, keep schemas controlled, and avoid unsupported tactical claims. Lead Developer review should occur before moving into Layer 3 modeling or AI interpretation.

End of General Developer 1 closeout document